



Substitution of molasses for corn grain at two levels of degradable protein. II. Effects on ruminal fermentation, digestion, and nitrogen metabolism

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ABSTRACT

Our objective was to evaluate cow N metabolism and ruminal measures with diets containing 3 different levels of molasses or finely ground dry corn grain with 2 levels of ruminally degradable protein (RDP). Twelve lactating, ruminally cannulated Holstein cows (parity 2.25 ± 0.62 ; 185 ± 56 DIM; 41.3 ± 6.3 kg of milk initially) were individually fed in an experiment designed as a split-plot, replicated 3×3 Latin square, where each period lasted 28 d. Six diets were formulated according to a 2×3 factorial arrangement of treatments, where 2 levels of RDP (+RDP and –RDP) were fed throughout the experiment as the whole plot, and 3 levels of molasses (0, 5.25, or 10.50% of dry matter replacing finely ground dry corn grain) were fed in sequences of the Latin squares. Dry matter intake did not differ by diet, although ash intake increased linearly with increasing molasses. Ruminal pH, organic acid concentration, and ammonia concentration were not affected by diet. Molar percentages of ruminal acetate decreased and butyrate increased linearly with increasing levels of molasses. Ruminal free amino acid concentration was greater for +RDP, whereas branched-chain volatile fatty acids declined linearly with increasing molasses. Rumen content mass, ruminal liquid, and ruminal acetate pool size was greater for –RDP, although ruminal lactate pool size tended to be greater with +RDP. Increased ruminal lactate when increasing molasses with high RDP should be explored further, to optimize microbial efficiency and rumen health. Total-tract apparent dry matter digestibility based on spot sampling was not affected by diet; however, ash digestibility increased linearly with increasing levels of molasses. Calculated urine output was greater for cows fed diets with increasing levels of molasses and for cows fed +RDP. Grams of N distributed to excretion pools were not different across diets, although, as molasses in-

creased, a lower proportion of N intake was excreted in urine. Overall, the results from this experiment showed that dairy cows used dietary carbohydrates differently during ruminal fermentation, with increasing molasses resulting in increased butyrate molar proportions at the expense of acetate. Additionally, RDP tended to modify the effects of carbohydrate fermentation, resulting in a tendency for increasing lactate molar pool size only in diets with greater RDP, although this did not ultimately affect ruminal pH.

Key words: molasses, rumen-degradable protein, nitrogen metabolism

INTRODUCTION

Carbohydrate and protein often comprise over 80% of feed DM in the diets of lactating dairy cows. These 2 distinct fractions can be further subdivided into various chemical fractions such as NDF, starch, sugar, and, with knowledge of ruminal degradation and passage rates, RDP and RUP (Higgs et al., 2015). Each of these dietary components is integrated by the ruminal microbial community to provide substrates for microbial growth, which in turn provides high-quality microbial protein (Roman-Garcia et al., 2016) and volatile fatty acids (Penner et al., 2009) for host metabolism. Ruminal protein degradation has been estimated to occur at different rates among different feedstuffs and protein fractions (Broderick et al., 1988; Van Amburgh et al., 2015). Likewise, different sources of carbohydrates are fermented at various rates (Hristov et al., 2005), where generally the rate of ruminal degradation increases from NDF to starch to sugars, although there is a wide range of fermentability of each of the carbohydrate sources based on storage, processing, chemical composition, ruminal adaptation, and other dietary factors (Weisbjerg et al., 1998; Oba and Allen, 2003).

Although carbohydrate degradation rate affects ruminal microbial utilization of carbohydrates, molecular composition of the nonstructural carbohydrate monomers, oligomers, or polymers can also affect ruminal utilization and end product formation (Gao and Oba, 2016; Hall and Weimer, 2016). Two of the main non-

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structural carbohydrates in the diets of dairy cows are starch and sugar. Exchanging sugar for starch has resulted in increased butyrate molar proportions in vitro when pH was maintained near neutrality (Strobel and Russell, 1986; Vallimont et al., 2004; Hoover et al., 2006). Additionally, increased dietary sugar has resulted in reduced molar proportions of acetate (Hoover et al., 2006) in vitro and decreased molar proportions of branched-chain VFA in vivo (Sannes et al., 2002), although other in vivo results on ruminal fermentation have been less consistent (Broderick and Radloff, 2004; Broderick et al., 2008; Hall et al., 2010).

Changing the level and fermentability of dietary carbohydrates and RDP have independently and sometimes interactively altered ruminal fermentation and microbial protein flow and efficiency (Stokes et al., 1991; Aldrich et al., 1993; Hall, 2013). In addition, the ruminal fermentation effects of replacing starch with sugar when diets contain different levels of RDP is unclear. Hristov et al. (2005) showed that ammonia utilization or production differed for cows provided glucose or starch intraruminally, although few carbohydrate source $\times$ RDP interactions were observed when studied in vivo (Hall et al., 2010; Sun et al., 2019). In light of these uncertainties, our objective was to evaluate cow N metabolism and ruminal measures with diets containing 3 different levels of molasses or finely ground dry corn grain with 2 levels of RDP. Our primary hypothesis was that, as carbohydrate fermentability was increased by feeding increasing levels of molasses, ruminal organic acid composition would shift, which would be indicated by increased butyrate molar proportion. Our secondary hypothesis was that ruminal responses to the changing proportions of molasses and ground corn would be modified by the degradability of dietary protein and result in decreased ruminal pH and increased ruminal lactate with greater RDP.

MATERIALS AND METHODS

To accomplish our objective, 12 ruminally cannulated, multiparous Holstein cows (parity 2.25 ± 0.62 ; 185 ± 56 DIM; 41.3 ± 6.3 kg of milk initially) were randomly allocated first to protein degradability level and then to carbohydrate treatment sequences in a split-plot, 3×3 Latin square design with a 2×3 factorial arrangement of treatments resulting in 36 cow $\times$ period observations. The whole-plot factor was the level of dietary RDP, which had 6 cows per RDP level (predicted positive and negative ruminal RDP balances, **+RDP** and **−RDP**) that did not change over to the other RDP level (**+RDP** cows: 182 ± 51 DIM, 722 ± 67 kg BW, 29.5 ± 3.2 kg/d DMI, 4.1 ± 0.3 daily

DMI as % of BW, 47.6 ± 5.3 kg/d milk yield; **−RDP** cows: 173 ± 37 DIM, 705 ± 64 kg BW, 28.4 ± 2.8 kg/d DMI, 4.0 ± 0.4 daily DMI as % of BW, 50.5 ± 9.0 kg/d milk yield; average $\pm$ SD at the beginning of the experiment; these values did not differ between RDP). The subplot factor was the sequences of the Latin squares in which cows changed among 3 diets with differing levels of molasses and dried ground corn. The Latin squares were arranged so that each molasses level followed every other molasses level an equal number of times. The experimental design and animal numbers were chosen based on experience from similar research conducted in our laboratory, availability of rumen-cannulated cows, and in order to balance the duration of the experiment with the length of lactation. Cows were housed in straw-bedded tiestalls and fed individually with ad libitum access to feed and water. Subsamples of feed offered and refused and forages were collected daily and composited by week; concentrates were collected once weekly.

Ingredient composition of feeds used to formulate treatment diets is shown in Table 1 (further details of feed composition are also provided in our companion paper, Hall and Zanton, 2022). Treatment diets were formulated with brown midrib corn silage, alfalfa silage, dried distillers grains with solubles, and vitamin and mineral premix (containing monensin; Table 2) at equal concentrations across dietary treatments. Other ingredients were varied to develop the treatment differences in formulated RDP and level of molasses. Specifically, solvent soybean meal (**SBM**) was replaced by expellers SBM (SoyPlus, Landus Cooperative), and a portion of dried ground corn was replaced with 5.25% or 10.5% (DM basis) molasses (Westway Feed Products). Treatment diets were formulated at a level of CP and RDP that was predicted to result in positive and negative ruminal RDP balances for **+RDP** and **−RDP**, respectively (based on the NRC, 2001, analysis detailed in Hall and Zanton, 2022). Amounts of feed offered and refused were weighed daily, and refusals were removed daily before morning feeding. All experimental procedures involving the use of animals were approved by the University of Wisconsin Institutional Animal Care and Use Committee (protocol no. A005043; Madison, WI).

Each period lasted 28 d, with the first 20 d for adaptation to dietary treatment and the last 8 d for sample collection. Diets were delivered to cows at approximately 0800 h daily throughout adaptation; on the day before initiating intensive sampling, cows were changed to $2\times$ feeding at 0800 and 2000 h in an attempt to more closely approximate a steady state. Milk weight was recorded at each of 3 daily milkings (approximately 0400, 1100, and 1830 h) every day of the

study, and milk was sampled at each milking on d 24 to 27. Body weight was measured after the third daily milking on d 24 to 27. On d 21 of each period, rumen contents of all cows were completely removed through the rumen cannula beginning at 1400 h, which was 6 h after feeding. Samples were removed to two tared 208-L garbage cans, with subsamples of rumen contents for subsequent analysis taken by placing every tenth handful of rumen contents into a tared 19-L pail. Rumen contents were weighed, subsampled by thoroughly mixing manually, and subsequently returned to the rumen of the same cow.

Feces and urine samples were taken as spot samples at intervals to represent every 4 h of the day (d25: 1200 h; d26: 0000 and 1600 h; d27: 0400, 0800, and 2000 h). Samples were collected as animals voluntarily defecated or urinated, when available; otherwise feces were removed by rectal sampling and urination was induced by stimulating the vulva. Urine samples were acidified immediately after collection by diluting 1 volume of urine with 4 volumes of 0.072 *N* H₂SO₄ and stored at −20°C until analysis. On d 25 to 28 of each period, samples of ruminal contents were taken at intervals to represent every 2 h of the day, to account for circadian variation (d25: 1200 and 1800 h; d26: 0000, 0600, 1000, 1600, and 2200 h; d27: 0400, 0800, 1400, and 2000 h; d28: 0200 h). Ruminal fluid was removed from 3 locations of the rumen (anterior dorsal, medial ventral, and posterior dorsal) using a straining sampling probe. Rumen fluid pH was measured immediately, and a sub-

sample was retained for further analysis, preserved with H₂SO₄, and stored at −20°C. At 1400 h on d 28 of each period, rumens were again emptied using the protocol described previously.

Analysis of samples proceeded similarly to what has been described previously (Paula et al., 2018) and was the only time during the experiment that blinding to treatment occurred. Briefly, milk samples were analyzed for composition using for fat, true protein, and MUN by infrared analysis (AgSource, Verona, WI) with a spectrum analyzer (FT6000; Foss North America Inc.). Concentrations and yields of fat, true protein, and MUN were calculated as weighted means based on the 3 milking yields on each day, and milk NE_L output was calculated as described in the NRC (2001). After drying, ingredients, TMR, and refusals were ground to pass a 1-mm screen (Wiley mill, Arthur H. Thomas). Samples were analyzed for DM, ash, OM, and total N using a combustion assay (Leco FP-2000 N Analyzer, Leco Instruments Inc.). Samples were analyzed for NDF after being treated with thermo-stable α-amylase and Na₂SO₃ using the Ankom²⁰⁰ Fiber Analyzer, using F57 bags with 25-μm pore size (**aNDF**; Schlau et al., 2021). Indigestible NDF was analyzed in the NDF residue remaining after 12-d in situ incubation in the rumen of 2 cows (25-mm porosity, 5 × 5 cm², 350-mg sample; Ankom Technology) as described for NDF. Residual NDF was ashed, and all NDF values are reported on an OM basis; potentially digestible NDF (**pdNDF**) is calculated as aNDF minus indigestible aNDF (**iNDF**).

Table 1. Chemical composition of feedstuffs collected during the last week of each period of the 3 × 3 Latin square with cannulated cows fed 2 levels of protein degradability and 3 levels of molasses

Item, ¹ % of DM, unless indicated	Corn silage	Alfalfa silage	Dried ground corn	Cane molasses	Soybean hulls	Solvent soybean meal	Expellers soybean meal	DDGS ²
DM, % AF	38.0	46.9	88.4	66.3	90.9	89.5	90.0	90.2
OM	96.6	88.9	98.6	83.9	94.9	92.4	93.2	94.5
Ash	3.4	11.1	1.4	16.1	5.1	7.6	6.8	5.5
NDF	32.9	35.4	8.3	—	66.2	8.1	17.2	30.1
pdNDF	27.0	18.2	8.2	—	64.0	8.3	16.5	29.2
iNDF	5.9	17.2	0.1	—	2.1	0.0	0.6	0.9
CP	7.8	23.2	8.5	7.8	11.9	52.4	47.4	34.2
NDICP	0.9	3.1	1.1	—	4.1	1.7	10.4	6.2
ADICP	0.4	1.2	0.3	—	1.0	1.2	1.3	2.5
Particle size ³								
AF TMR > 19mm	4.8	4.0						
19 mm > AF TMR > 8 mm	75.3	69.3						
8 mm > AF TMR > 4 mm	16.8	16.9						
4 mm > AF TMR > 1.18 mm	2.9	7.5						
1.18 mm > AF TMR	0.2	2.3						
Mean, mm	10.7	8.9						
SD, mm	1.1	1.5						

¹AF = as fed; NDF = NDF, ash-corrected; pdNDF = potentially digestible NDF; iNDF = indigestible NDF; NDICP = neutral detergent insoluble CP; ADICP = acid detergent insoluble CP.

²DDGS = dried distillers grains with solubles.

³Particle size determined using the Penn State Particle Size Separator on forage samples collected during the last week of each period of the 3 × 3 Latin square. Presented on an AF basis.

Table 2. Ingredients and chemical composition of mixed diets fed during the last week of each period of the 3 × 3 Latin square with cannulated cows fed 2 levels of protein degradability and 3 levels of molasses¹

Molasses, % of DM	+RDP			-RDP		
	0	5.25	10.5	0	5.25	10.5
Ingredient, ² % of DM, unless indicated						
BMR corn silage	34.6	34.6	34.4	34.7	34.5	34.6
Alfalfa silage	21.2	21.1	21.0	21.2	21.1	21.2
Dried ground corn	18.6	14.3	10.0	18.7	14.5	10.0
Cane molasses	0.0	5.1	10.4	0.0	5.2	10.3
Soybean hulls	7.5	6.8	6.0	6.5	5.9	5.3
Solvent soybean meal	13.0	13.2	13.5	6.4	6.3	6.4
Expellers soybean meal	0.0	0.0	0.0	7.4	7.5	7.7
Dried distillers grains with solubles	2.4	2.0	2.0	2.1	2.0	2.0
Limestone	0.27	0.14	0.00	0.27	0.14	0.00
KMgS	0.27	0.14	0.00	0.27	0.14	0.00
Vitamin and mineral premix ³	2.6	2.6	2.6	2.6	2.6	2.6
Chemical composition, % of DM, unless indicated						
DM, % AF	53.9	53.4	52.9	53.9	53.4	52.8
OM	91.6	91.1	90.6	91.6	91.1	90.6
Ash	8.4	8.9	9.4	8.4	8.9	9.4
aNDF	26.9	26.0	25.0	27.0	26.1	25.3
pdNDF	21.1	20.2	19.2	21.1	20.3	19.5
iNDF	5.8	5.8	5.8	5.9	5.8	5.9
CP	17.4	17.4	17.4	17.4	17.3	17.3
NDICP	1.83	1.74	1.66	2.44	2.37	2.30
ADICP	0.70	0.68	0.66	0.70	0.68	0.66
TMR particle size ⁴						
AF TMR > 19 mm	1.92	2.34	1.75	1.77	1.87	2.03
19 mm > AF TMR > 8 mm	35.6	36.3	38.7	37.3	36.8	39.7
8 mm > AF TMR > 4 mm	10.5	11.6	11.7	11.2	11.4	12.1
4 mm > AF TMR > 1.18 mm	34.4	33.9	33.9	33.5	33.3	32.2
1.18 mm > AF TMR	17.6	15.8	14.0	16.3	16.6	14.0
Mean, mm	4.0	4.2	4.4	4.2	4.1	4.5
SD, mm	1.61	1.60	1.57	1.60	1.60	1.58

¹Chemical composition is based on actual feeding amounts and chemical analysis of the feed ingredients. +RDP and -RDP = positive and negative ruminal RDP balances, respectively.

²AF = as fed; NDF = NDF, ash-corrected; pdNDF = potentially digestible NDF; iNDF = indigestible NDF; NDICP = neutral detergent insoluble CP; ADICP = acid detergent insoluble CP.

³Containing (DM basis): 16.8% Ca, 4.64% Mg, 0.6% K, 15.8% Na, 7.1% Cl, 1.0% S, 45 mg/kg Co, 554 mg/kg Cu, 64 mg/kg I, 778 mg/kg Fe, 2,601 mg/kg Mn, 15.6 mg/kg Se, 2,808 mg/kg Zn, 311 kIU/kg vitamin A, 62 kIU/kg vitamin D, 1,453 IU/kg vitamin E, 523 mg/kg Rumensin (Elanco).

⁴TMR particle size determined using the Penn State Particle Size Separator on TMR samples collected during the last week of each period of the 3 × 3 Latin square. Presented on an AF basis.

Sorting was evaluated for pdNDF, iNDF, and CP by modification of the approach described by Leonardi and Armentano (2003) for particle size. In this method, actual intake (ort-corrected) is expressed relative to predicted intake (concentration of nutrient in DM offered times DMI), this is then centered at 0 by subtracting 1, and the existence of sorting is tested as a difference from 0, where values greater than 0 indicate sorting for the nutrient and less than 0 indicate sorting against the nutrient.

Fecal samples were dried in a forced-draft oven (55°C; 72 h) and ground to pass a 1-mm screen (Wiley mill). Equal DM from each fecal subsample was combined to obtain one composite sample for each cow in each period. Fecal samples were analyzed for total DM, ash, OM, N, NDF, and iNDF as described for feed analysis.

Indigestible aNDF was used as the internal marker for estimating fecal output and digestibility, as reported previously (Zanton, 2019). Urine samples were thawed at room temperature and analyzed for creatinine using a picric acid method adapted to flow-injection analysis (Lachat Quik-Chem 8000 FIA; Lachat Instruments), total N using a Dumas combustion analysis (Leco FP-2000 N Analyzer; Leco Instruments Inc.), allantoin, uric acid, and urea using colorimetric methods (Paula et al., 2018). Daily urine volume was calculated based on individual BW and using creatinine excretion rate of 29 mg/kg of BW (Valadares et al., 1999). Urinary urea N, total N, total purine derivatives, and allantoin plus uric acid were calculated based on their individual daily concentration multiplied by estimated daily urine volume.

Ruminal samples were thawed at room temperature and centrifuged ($15,300 \times g$ for 20 min at 4°C), and flow-injection analyses (Lachat Quik-Chem 8000 FIA; Lachat Instruments) were applied to supernatants to determine ammonia, using a phenol-hypochlorite method (Lachat Method 18-107-06-1-A; Lachat), and total AA as leucine equivalents. Ruminal organic acid concentrations were determined using HPLC (Weimer et al., 1991).

Evacuated rumen content subsamples were dried (55°C for 72 h) in a forced-air oven, and content mass lost during drying is defined as rumen liquid. Evacuated contents were also squeezed through 4 layers of cheesecloth, with the mass passing through the cheesecloth defined as rumen fluid. Rumen content pool sizes are calculated by multiplying the proportions of fluids and solids or liquid and DM by the mass of evacuated contents. Likewise, rumen fermentation parameter pool sizes are calculated by multiplying organic acid or nitrogenous concentrations determined at 1400 h in rumen fluid by the rumen liquid mass.

Data were analyzed as a split-plot Latin square design with fixed effects of period, protein degradability, molasses level, and the interaction between protein degradability and molasses level. Effect of previous treatment (carryover) was also evaluated for some responses as a fixed effect. Cow within protein degradability was the random effect, which accounts for the lower level of replication across protein degradability and results in cow(protein degradability) as the whole-plot error term and 10 degrees of freedom (df) for the test of protein degradability effects. Residual error was used in tests of carbohydrate and interactions with df = 18. All analyses were conducted in SAS using the MIXED procedure (SAS Institute, 2013). Normality was assessed using normal probability plots, and if assumptions of normality were violated, an appropriate transformation was explored. For instances in which repeated measures were made within a period, autocorrelation between time points was modeled using the first-order autoregressive function. Orthogonal contrasts of interest were the main effects of protein degradability, linear and quadratic effects of carbohydrate type, and the linear and quadratic interaction between protein degradability and carbohydrate type. Sorting was evaluated by testing the least squares means difference from 0 through the use of a *t*-test. Significance was declared when $P \leq 0.05$ and trending toward significance when $0.05 < P \leq 0.10$.

RESULTS

Ash concentration numerically increased and pdNDF numerically declined with increasing dietary molasses

concentration (Table 2), although iNDF concentration was not different across diets. These responses in pdNDF and iNDF resulted in total aNDF concentration numerically decreasing with increased molasses concentration. Crude protein content did not differ across diets, but neutral detergent insoluble CP concentration was greater in the -RDP diets, as planned. Dietary mean particle size (as-fed basis) also increased with increasing molasses concentration. Concentration of iNDF changed between feed offered and refused as cows selected for iNDF across all diets, but diet did not affect sorting ($P > 0.067$) when expressed as sorting index (Table 3).

Dry matter intake was not affected by diet ($P > 0.40$; Table 4), although intake of ash increased linearly ($P < 0.001$) and intake of pdNDF tended to decline ($P = 0.054$) with increasing molasses and quadratically interact with RDP ($P = 0.052$). Body weight ($P = 0.033$) and BW change ($P = 0.002$) increased linearly with increasing molasses (Table 4), and productive performance was not affected by diet ($P > 0.083$), except for a quadratic interaction ($P = 0.047$) in milk fat yield where the quadratic effect increased at intermediate molasses with +RDP and decreased with -RDP. However, these results must be treated with caution because of the short periods used in this 3×3 Latin square design. More extensive discussion of lactation performance is provided in our companion manuscript (Hall and Zanton, 2022).

Average ruminal pH, organic acid concentration, and ammonia concentration were not affected by diet ($P > 0.10$; Table 5) and followed a similar temporal pattern (Figure 1). The time that ruminal pH was less than 6 tended ($P = 0.069$) to increase with increasing molasses; however, pH in this study was not at a level that would generally be considered concerning (average pH = 6.26). Ruminal acetate molar percent decreased linearly ($P < 0.050$) and butyrate molar percent increased linearly ($P < 0.001$) with increasing levels of molasses (Table 5). Acetate and butyrate molar proportions were also affected by a time after feeding $\times$ molasses level interaction ($P = 0.052$ and $P < 0.001$, respectively; Figure 2) where higher levels of molasses resulted in a larger reduction in acetate molar percent and a larger increase in butyrate molar percent after feeding. Lactate concentration was increased ($P = 0.020$) and maximum lactate tended to increase ($P = 0.072$) quadratically where ruminal lactate concentration was minimized at the intermediate level of molasses. Maximum lactate also tended to be affected by a protein $\times$ molasses quadratic interaction ($P = 0.082$) where maximum lactate increased with increasing molasses in +RDP, but 5.25% molasses resulted in the lowest maximum lactate when fed with -RDP. Ruminal free amino acid concentra-

tion was greater ($P = 0.020$) for diets formulated with +RDP and was lowest at 5.25% molasses (Table 5). Branched-chain VFA declined linearly ($P < 0.001$) with increasing molasses.

Rumen content mass was greater for diets formulated with -RDP ($P = 0.030$; Table 6). This effect was due to significantly greater quantities of rumen liquid ($P = 0.018$), although rumen DM mass was numerically greater as well. Ruminal total organic acid pool size did not differ across diets ($P > 0.10$). Ruminal acetate pool size was greater ($P = 0.044$) for cows fed diets formulated with -RDP, and butyrate pool size increased ($P = 0.041$) as cows were fed diets with increasing concentrations of molasses. Ruminal lactate pool size tended to be greater ($P = 0.054$) in diets formulated with +RDP; however, the distribution of the data precluded a more rigorous statistical analysis (Figure 3), as many observations were 0. However, within +RDP, increasing the level of molasses resulted in a numerically increased median, average, and maximum lactate pool size. Ammonia and free AA pool sizes were not affected ($P > 0.10$) by diets, and, consistent with branched-chain VFA (BCVFA) concentration, BCVFA pool size declined ($P = 0.004$) with increasing levels of molasses.

Total-tract apparent DM, OM, aNDF, and pdNDF digestibilities were not affected by diet ($P > 0.10$; Table 7); however, ash digestibility increased linearly ($P = 0.017$) with increasing levels of molasses. Crude protein digestibility trended lower at intermediate levels of molasses than at 0 or 10.5% molasses ($P = 0.077$; quadratic tendency). Calculated urine output was greater ($P < 0.001$) for cows fed diets with increasing levels of molasses and for cows fed diets formulated with +RDP ($P = 0.050$; Table 8). Grams of N distributed to excretion pools were not different ($P > 0.10$) across diets, although a lower proportion of N intake was excreted in urine as molasses increased ($P = 0.030$ linear; $P = 0.038$ quadratic). Urinary urea N (UUN) concentration was lower ($P < 0.001$) with increasing levels of molasses, although UUN and purine derivative excretion were not different due to diet ($P > 0.10$).

DISCUSSION

In our study, we were interested in understanding ruminal fermentation and pool sizes, total-tract apparent digestibility, and N partitioning responses to increasing dietary sugar through molasses supplementation,

Table 3. Total mixed ration offered, refused, and consumed, pdNDF and iNDF analyses, and sorting indices (SI) for cannulated cows fed 2 levels of protein degradability and 3 levels of molasses¹

Item, % of DM, unless noted otherwise ²	Protein ³	Molasses, % of DM				P-value ⁴				
		0	5.25	10.5	SEM	Protein	Carbohydrate		Interaction	
							L	Q	L	Q
pdNDF										
Offered	+RDP	23.0	21.0	21.0	0.35	0.002	0.001	0.993	0.559	0.002
	-RDP	20.9	21.2	19.3						
Refused	+RDP	22.2	20.9	19.3	0.47	0.333	0.001	0.580	0.034	0.349
	-RDP	20.7	20.8	20.0						
Consumed	+RDP	23.1	21.0	21.1	0.36	0.002	0.001	0.978	0.602	0.002
	-RDP	20.9	21.2	19.3						
SI ⁵	+RDP	0.27	0.06	0.29	0.162	0.276	0.699	0.639	0.590	0.032
	-RDP	-0.03	0.25	-0.16						
iNDF										
Offered	+RDP	6.8	6.6	6.5	0.22	0.023	0.401	0.305	0.664	0.280
	-RDP	7.1	7.5	7.0						
Refused	+RDP	5.6	5.5	5.8	0.36	0.646	0.724	0.331	0.167	0.768
	-RDP	5.6	5.4	5.3						
Consumed	+RDP	6.9	6.7	6.6	0.23	0.018	0.420	0.252	0.586	0.246
	-RDP	7.2	7.6	7.2						
SI ⁶	+RDP	1.07	1.08	0.70	0.449	0.132	0.934	0.067	0.162	0.309
	-RDP	1.51	2.30	1.84						

¹Sorting index = actual intake with sorting/predicted intake without sorting -1. Values greater than 0 reflect selective consumption; values less than 0 reflect selective refusal. pdNDF = potentially digestible NDF; iNDF = indigestible NDF.

²Analysis of offered and refused samples collected during the last week of each period of the 3 × 3 Latin square.

³+RDP and -RDP = positive and negative ruminal RDP balances, respectively.

⁴L = linear; Q = quadratic.

⁵pdNDF SI did not differ from 0 for any diet.

⁶iNDF SI differed from 0 for all diets except for RDP, 10.5% molasses ($P = 0.135$).

as well as whether RDP level had an effect on these responses. To test these dietary alterations, we used a split-plot Latin square design in which cows did not change between RDP levels but received all levels of molasses in accordance with the Latin square sequences. Due to this design, RDP responses could be influenced by the chance assignment of cows to whole plots and unexpected differences among these cows; however, the substitution of corn for molasses and the interaction of RDP and molasses level would not be affected. Substitution studies of this nature require the removal of another ingredient to balance the treatment diets to 100%. For this study, we decided to remove finely ground dried corn and substitute molasses into the

treatment diets, because it would allow a comparison of 2 frequently fed, readily fermentable carbohydrate sources with different rates of degradation and chemical constituents. Likewise, diets were kept isonitrogenous by removing a portion of solvent SBM and soybean hulls and substituting expellers SBM. In contrast to studies in which purified nutrient sources (starch and sucrose) were exchanged (Broderick et al., 2008; Sun et al., 2019), these feedstuff substitution decisions have effects beyond our primary nutrients of interest. For example, pdNDF declined with increasing molasses content due to the removal of pdNDF from corn. The most prominent noncarbohydrate chemical constituent that differed between molasses and corn was ash concentra-

Table 4. Intake and production responses for cannulated cows fed 2 levels of protein degradability and 3 levels of molasses

Item, kg/d unless noted otherwise ¹	Protein ²					<i>P</i> -value ³				
		Molasses, % of DM			SEM	Protein	Carbohydrate		Interaction	
		0	5.25	10.5			L	Q	L	Q
DM intake	+RDP	29.0	29.9	30.2	1.37	0.975	0.440	0.594	0.558	0.848
	−RDP	29.4	30.0	29.6						
OM intake	+RDP	26.7	27.3	27.5	1.26	0.940	0.665	0.623	0.606	0.835
	−RDP	26.9	27.4	26.9						
Ash	+RDP	2.3	2.6	2.8	0.11	0.643	0.001	0.358	0.209	0.990
	−RDP	2.5	2.7	2.7						
NDF	+RDP	8.7	8.3	8.4	0.41	0.715	0.142	0.478	0.733	0.089
	−RDP	8.3	8.6	7.8						
pdNDF	+RDP	6.7	6.3	6.4	0.30	0.330	0.054	0.616	0.712	0.052
	−RDP	6.2	6.4	5.7						
iNDF	+RDP	2.0	2.0	2.0	0.14	0.279	0.873	0.356	0.832	0.412
	−RDP	2.1	2.3	2.1						
N	+RDP	0.81	0.83	0.84	0.038	0.896	0.541	0.621	0.487	0.844
	−RDP	0.82	0.83	0.82						
BW, kg	+RDP	731	730	740	28.5	0.862	0.033	0.187	0.416	0.434
	−RDP	725	726	729						
BW change ⁴	+RDP	−0.31	0.13	0.39	0.142	0.419	0.002	0.733	0.137	0.734
	−RDP	0.03	0.17	0.30						
Milk	+RDP	40.9	43.0	42.3	2.65	0.920	0.562	0.882	0.635	0.267
	−RDP	42.7	41.7	42.9						
Milk NE _L , ⁵ Mcal/d	+RDP	30.3	32.0	31.1	1.83	0.873	0.342	0.900	0.940	0.129
	−RDP	30.6	29.9	31.6						
Fat	+RDP	1.64	1.76	1.69	0.099	0.422	0.193	0.971	0.681	0.047
	−RDP	1.57	1.52	1.66						
Protein ⁶	+RDP	1.37	1.44	1.42	0.079	0.458	0.403	0.906	0.834	0.304
	−RDP	1.33	1.31	1.36						
Fat, %	+RDP	4.05	4.10	4.04	0.145	0.155	0.170	0.554	0.135	0.133
	−RDP	3.70	3.68	3.91						
Protein, ⁶ %	+RDP	3.36	3.33	3.36	0.070	0.083	0.626	0.690	0.779	0.537
	−RDP	3.15	3.17	3.18						
MUN, mg/dL	+RDP	12.4	11.8	12.3	0.38	0.666	0.286	0.376	0.471	0.491
	−RDP	12.7	12.3	12.0						

¹Intake based on quantity of feed offered and refused and analysis of offered and refused samples collected during the last week of each period of the 3 × 3 Latin square except for N intake, which is based on ingredient chemical composition and feeding amount. pdNDF = potentially digestible NDF; iNDF = indigestible NDF.

²+RDP and −RDP = positive and negative ruminal RDP balances, respectively.

³L = linear; Q = quadratic.

⁴Body weight change over each 28-d period, expressed as kg/d.

⁵Net energy of lactation predicted from milk components (NRC, 2001).

⁶True protein.

Table 5. Average ruminal fermentation responses for cannulated cows fed 2 levels of protein degradability and 3 levels of molasses

Item ¹	Protein ²	P-value ³								
		Molasses, % of DM			SEM	Protein	Carbohydrate		Interaction	
		0	5.25	10.5			L	Q	L	Q
pH	+RDP	6.31	6.27	6.26	0.068	0.640	0.492	0.563	0.321	0.876
	–RDP	6.24	6.22	6.25						
pH <6.0, hr	+RDP	3.2	4.7	4.5	1.70	0.701	0.069	0.920	0.583	0.289
	–RDP	4.1	4.3	6.5						
Organic acid, ⁴ mM	+RDP	116	117	119	3.6	0.658	0.685	0.751	0.228	0.469
	–RDP	120	121	119						
Organic acid C, mM	+RDP	288	291	297	10.2	0.779	0.309	0.771	0.306	0.535
	–RDP	295	299	295						
Acetate, mol/100 mol	+RDP	67.0	66.4	66.1	1.12	0.545	0.050	0.915	0.920	0.731
	–RDP	67.8	67.5	67.0						
Propionate, mol/100 mol	+RDP	20.7	20.6	20.0	1.01	0.541	0.582	0.942	0.383	0.333
	–RDP	19.6	19.3	19.7						
Butyrate, mol/100 mol	+RDP	11.2	11.8	12.6	0.26	0.988	0.001	0.785	0.191	0.474
	–RDP	11.4	12.0	12.1						
Valerate, mol/100 mol	+RDP	1.3	1.3	1.3	0.046	0.215	0.194	0.816	0.303	0.493
	–RDP	1.2	1.2	1.2						
Lactate, ⁵ mM	+RDP	0.39	0.36	0.63	—	0.107	0.212	0.020	0.922	0.168
	–RDP	0.27	0.10	0.41						
Maximum lactate, ⁵ mM	+RDP	2.43	3.14	4.20	—	0.143	0.254	0.072	0.660	0.082
	–RDP	2.42	0.92	3.09						
Ammonia, mM	+RDP	5.82	5.59	6.14	0.389	0.625	0.351	0.111	0.447	0.446
	–RDP	5.63	5.50	5.66						
Free AA, mM	+RDP	1.98	1.79	1.96	0.099	0.020	0.267	0.023	0.171	0.670
	–RDP	1.54	1.51	1.72						
BCVFA, ⁶ mM	+RDP	2.53	2.24	2.12	0.126	0.775	0.001	0.354	0.636	0.559
	–RDP	2.53	2.33	2.18						

¹Samples were collected to represent every 2 h.²+RDP and –RDP = positive and negative ruminal RDP balances, respectively.³L = linear; Q = quadratic.⁴Sum of acetate, propionate, butyrate, valerate, and lactate.⁵Lactate values are back-transformed from log-analyzed values.⁶BCVFA = branched-chain VFA, sum of isovalerate and isobutyrate.

tion, which was over 16% in molasses and under 1.5% in corn. This difference between ingredients resulted in ash and OM content changing linearly as molasses inclusion increased, with a maximum of approximately 1 percentage-unit difference between the 0% and 10.5% molasses diets.

Frequently, when cows have molasses substituted into their diets, DMI changes directly with the level of molasses (Broderick and Radloff, 2004, Trial 1; Torres et al., 2021) and with changing levels of sucrose in exchange for starch (Broderick et al., 2008; Sun et al., 2019). In contrast, DMI changes have not been consistently associated with dietary RDP content alterations when replacing RUP protein sources (Santos et al., 1998). In this experiment, when cows were fed increasing levels of molasses substituted for finely ground dried corn in a Latin square design and different levels of RDP, DMI was not different across diets and was quite high, averaging approximately 29.7 kg/d. Considering also the high BW of these cows, daily DMI as a percentage of BW averaged 4.08% and was not different across treat-

ment diets even though BW increased with increasing molasses.

Increasing BW and BW change with increasing levels of molasses was paralleled by the numerical increases in N balance that was observed as molasses content of the diet increased. Sun et al. (2019) also showed large increases in BW change with increasing dietary sucrose; however, other dose response studies with increasing molasses (Broderick and Radloff, 2004) or with sucrose (Broderick et al., 2008) did not show differences in BW change. A recent meta-analysis (Torres et al., 2021) demonstrated that increased molasses inclusion significantly reduced BW gain compared with control and found that this response was most evident when the basal diets contained between 60 and 80% concentrate.

Ultimately, the BW changes that were observed in response to dietary molasses in this changeover experiment with 28-d periods were not replicated in the concurrent longitudinal study reported in our companion paper (Hall and Zanton, 2022). Cows in this study were in later lactation than those in our companion

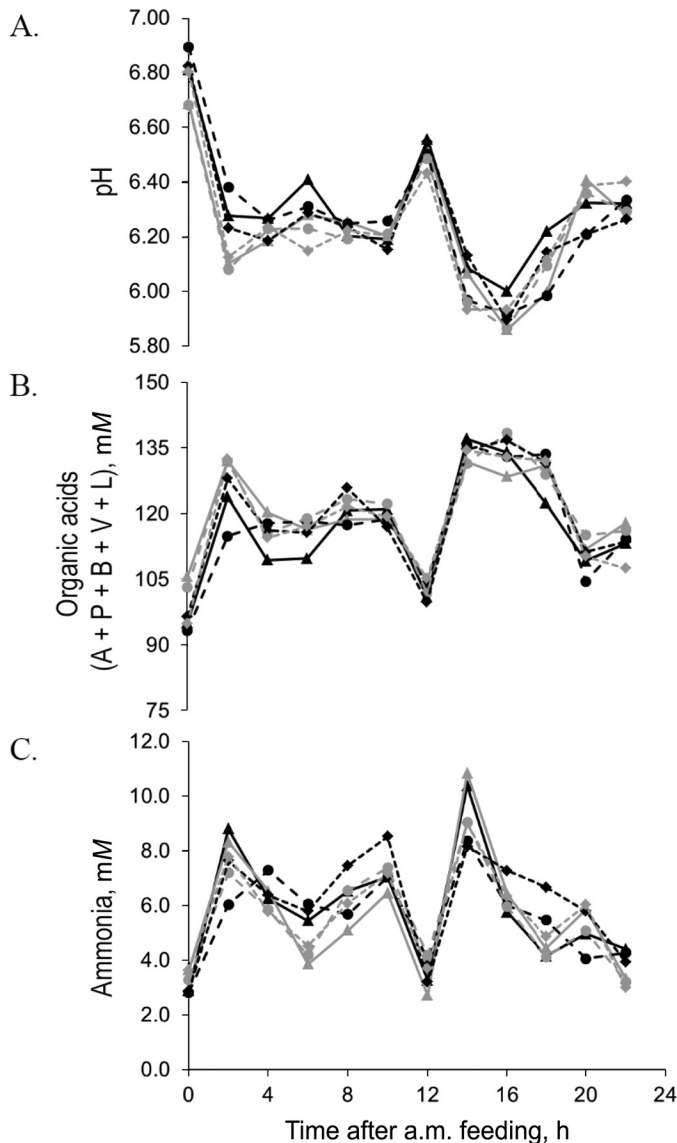


Figure 1. Ruminal fermentation profiles for (A) rumen pH, (B) organic acid concentration (mM; A = acetate, P = propionate, B = butyrate, V = valerate, L = lactate), and (C) ammonia concentration (mM) when cows were offered diets with positive (black symbols and lines) or negative ruminal RDP balance (gray symbols and lines) in combination with 0% molasses (triangles, solid lines), 5.25% molasses (circles, long-dashed lines), or 10.5% molasses (diamonds, short-dashed lines) at time 0 and 12 h.

study, which may affect the productivity and nutrient partitioning responses to these dietary treatments. However, this discrepancy may be more related to the diet changes between periods than to the diets alone, although the effect of previous dietary treatment carryover was not significant ($P > 0.80$) and rumen fill was not affected by molasses level in the diet ($P > 0.20$). A previous study determined that changeover designs did not reflect differences in measures related

to nutrient partitioning (BW and residual or retained N) compared with longitudinal studies when dietary protein levels differed (Zanton, 2019). More broadly, in addition to BW and BW change, DMI and productivity trait responses were also different between this study

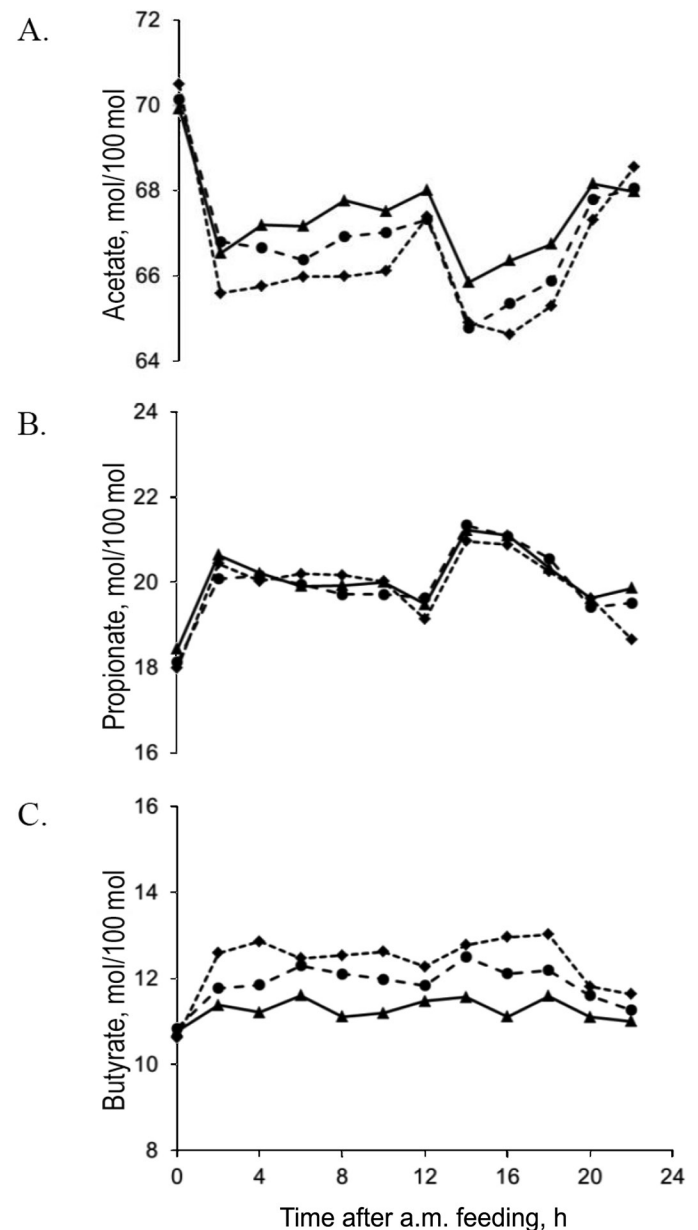


Figure 2. Ruminal fermentation profiles for (A) acetate, (B) propionate, and (C) butyrate (mol/100 mol) when cows were offered diets with 0% molasses (triangles, solid lines), 5.25% molasses (circles, long-dashed lines), or 10.5% molasses (diamonds, short-dashed lines) at time 0 and 12 h. Carbohydrate source and level significantly ($P < 0.001$) interacted or tended ($P < 0.06$) to interact with time for butyrate and acetate, respectively. Protein source or a protein $\times$ carbohydrate interaction did not interact with time for rumen VFA molar percentage profile ($P > 0.10$).

Table 6. Ruminal pool sizes for cannulated cows fed 2 levels of protein degradability and 3 levels of molasses¹

Item	Protein ²					P-value ³				
		Molasses, % of DM			SEM	Protein	Carbohydrate		Interaction	
		0	5.25	10.5			L	Q	L	Q
Rumen content mass, kg of wet weight	+RDP	91	92	94	4.6	0.030	0.202	0.985	0.917	0.790
	–RDP	106	108	109						
Rumen fluid, ⁴ kg of wet weight	+RDP	44	42	45	3.0	0.004	0.368	0.885	0.600	0.434
	–RDP	51	55	55						
Rumen solid, ⁴ kg of wet weight	+RDP	47	50	49	3.7	0.307	0.856	0.806	0.498	0.379
	–RDP	55	53	54						
Rumen liquid, ⁵ kg	+RDP	77	78	80	3.8	0.018	0.152	0.967	0.843	0.676
	–RDP	90	92	93						
Rumen DM, ⁵ kg	+RDP	14	14	14	1.0	0.234	0.817	0.786	0.714	0.574
	–RDP	16	16	16						
Rumen DM, ⁵ %	+RDP	15.3	15.4	14.8	0.52	0.389	0.173	0.813	0.795	0.485
	–RDP	15.1	14.6	14.4						
Fermentation pools ⁶										
Organic acid, mol	+RDP	8.5	9.3	9.3	0.75	0.106	0.133	0.319	0.697	0.905
	–RDP	10.4	11.0	10.9						
Organic acid C, mol	+RDP	21.2	23.3	23.4	1.99	0.138	0.110	0.325	0.770	0.905
	–RDP	25.7	27.3	27.2						
Acetate, mol	+RDP	5.63	6.06	6.05	0.423	0.044	0.249	0.272	0.430	0.939
	–RDP	7.07	7.31	7.15						
Propionate, mol	+RDP	1.78	1.91	1.88	0.223	0.343	0.178	0.551	0.594	0.851
	–RDP	2.01	2.17	2.24						
Butyrate, mol	+RDP	1.00	1.16	1.17	0.117	0.202	0.041	0.265	0.852	0.924
	–RDP	1.21	1.35	1.36						
Valerate, mol	+RDP	0.12	0.13	0.13	0.015	0.607	0.261	0.313	0.871	0.819
	–RDP	0.13	0.15	0.14						
Lactate, ⁷ mmol	+RDP	8	49	75	—	0.054	0.213	0.883	0.238	0.854
	–RDP	0	0	0						
Ammonia, mmol	+RDP	431	486	469	95.0	0.493	0.408	0.686	0.794	0.831
	–RDP	343	390	415						
Free AA, mmol	+RDP	119	150	149	21.6	0.338	0.107	0.795	0.887	0.490
	–RDP	103	114	139						
BCVFA, ⁸ mmol	+RDP	193	183	165	20.3	0.406	0.004	0.491	0.609	0.834
	–RDP	220	209	181						

¹Rumen contents were emptied before and after the last week of each period to determine rumen pool sizes. Rumen contents were emptied at 1400 h (midway between the 2 daily feedings; 6 h after feeding) at each occurrence.

²+RDP and –RDP = positive and negative ruminal RDP balances, respectively.

³L = linear; Q = quadratic.

⁴Rumen fluid that was squeezed through or solids retained on 4 layers of cheesecloth.

⁵Rumen contents lost on drying (liquid) or DM retained after drying.

⁶Pool sizes are the product of the mass of material lost on drying and the rumen fluid chemical analysis on rumen fluid samples collected during the frequent sampling.

⁷Lactate values are back-transformed from log-analyzed values.

⁸BCVFA = branched-chain VFA.

and the concurrently conducted longitudinal study. These differences could be related to the smaller study size, experimental design, or different experimental protocols, although these responses were also not significantly affected by a previous treatment carryover effect. Huhtanen and Hetta (2012) evaluated the effect of experimental design on treatment responses to several different dietary changes and concluded that, generally, responses to dietary changes were similarly predicted for both continuous and changeover studies. An important exception to this conclusion was when differences

between dietary responses were large, which was also the conclusion reached in a similar meta-analysis conducted when dietary CP levels were changed (Zanton, 2016), although protein degradability was not changed between periods in this study. The companion continuous study, evaluating production responses, resulted in relatively large differences in DMI and production; for this reason caution should be exercised in interpreting production responses in this study. Nevertheless, ruminal, digestibility, and N responses will be discussed with respect to direct effects of dietary treatments.

Table 7. Total-tract apparent nutrient digestibility for cannulated cows fed 2 levels of protein degradability and 3 levels of molasses

Apparent total-tract digestibility, ¹ %	Protein ²	Molasses, % of DM				P-value ³				
						Carbohydrate		Interaction		
		0	5.25	10.5	SEM	Protein	L	Q	L	Q
DM	+RDP	69.5	69.3	70.7	1.38	0.814	0.732	0.135	0.539	0.410
	−RDP	70.5	67.8	70.2						
OM	+RDP	71.5	71.2	72.2	1.28	0.679	0.940	0.163	0.600	0.464
	−RDP	72.0	69.7	71.5						
Ash	+RDP	45.0	51.2	56.5	2.65	0.269	0.017	0.168	0.099	0.123
	−RDP	54.6	48.8	56.9						
NDF	+RDP	51.8	49.5	50.3	2.24	0.553	0.271	0.588	0.754	0.661
	−RDP	50.5	49.0	47.8						
pdNDF	+RDP	67.2	65.2	65.5	2.65	0.787	0.296	0.820	0.763	0.688
	−RDP	68.2	67.0	65.1						
CP	+RDP	68.7	67.8	68.7	1.49	0.305	0.499	0.077	0.523	0.289
	−RDP	69.1	64.5	67.2						

¹Based on indigestible aNDF (NDF after being treated with thermostable α -amylase and Na_2SO_3) as an internal marker, using intake based on quantity of feed offered and refused and analysis of offered and refused samples collected during the last week of each period of the 3×3 Latin square, except for CP intake, which is based on ingredient chemical composition and feeding amount. pdNDF = potentially digestible NDF.

²+RDP and −RDP = positive and negative ruminal RDP balances, respectively.

³L = linear; Q = quadratic.

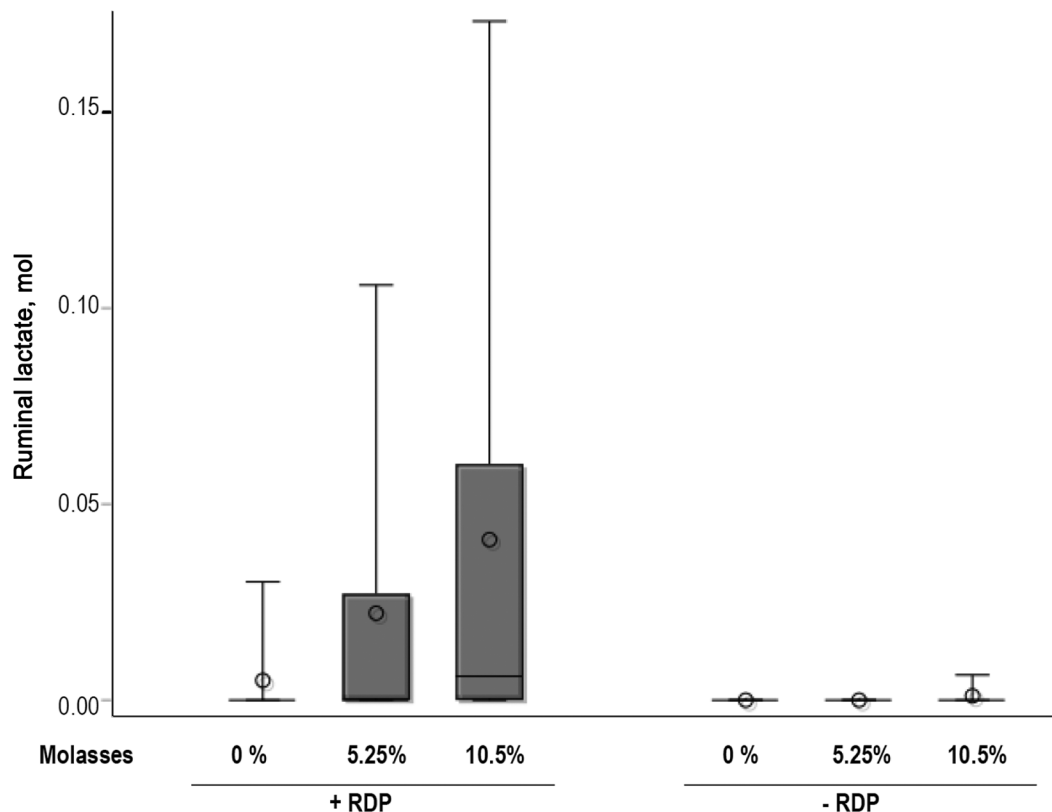


Figure 3. Boxplot of ruminal lactate pool size when cows were offered diets with positive or negative ruminal RDP balances (+RDP or −RDP, respectively) and 0% molasses, 5.25% molasses, or 10.5% molasses, based on rumen emptying 6 h after morning feeding. Upper bars (whiskers) represent maximum observed values; box is upper-bounded by the 75th percentile; circle is the mean value; line is the median value.

The results of our study support our hypothesis that ruminal organic acid composition shifts when carbohydrate fermentability was increased by increasing molasses and decreasing corn. This was observed in increasing butyrate and reduced acetate molar proportions as molasses increased. Reports in the literature vary regarding the effects on ruminal organic acid concentrations of feeding molasses or sucrose to ruminants (Oba, 2011; Torres et al., 2021). Under in vitro conditions, ruminal butyrate concentration is often increased (Strobel and Russell, 1986; Vallimont et al., 2004). In vivo responses to exchanging sucrose for starch sources have been much more variable, with some studies showing increased butyrate concentration (Hristov et al., 2005; Hall et al.,

2010; Sun et al., 2019) but others showing no change (Broderick and Radloff, 2004; Broderick et al., 2008). Torres et al. (2021) did not find a significant relationship between molasses supplementation and ruminal butyrate concentration, but acetate concentration significantly decreased in their meta-analysis. Strobel and Russell (1986) observed that the butyrate concentration response to sucrose compared with starch fermentation depended upon the pH present in the fermentation medium, with butyrate concentration increasing when pH was at 6.7, but unchanged when final pH was 5.5 to 5.8. In this in vitro study, lactate concentrations were increased independently by sucrose substrate and reduced pH, and acetate concentration was affected only

Table 8. Nitrogen distribution for cannulated cows fed 2 levels of protein degradability and 3 levels of molasses

Apparent N distribution, g/d	Protein ¹	Molasses, % of DM				P-value ²				
					SEM	Protein	Carbohydrate		Interaction	
		0	5.25	10.5			L	Q	L	Q
Intake ³	+RDP	810	832	842	38.2	0.896	0.541	0.621	0.487	0.844
	−RDP	818	831	816						
Digested ⁴	+RDP	556	563	577	22.8	0.528	0.818	0.412	0.299	0.538
	−RDP	562	534	548						
Feces	+RDP	254	269	265	21.3	0.657	0.504	0.128	0.970	0.363
	−RDP	256	297	268						
Urine ⁵	+RDP	339	307	323	16.8	0.276	0.204	0.138	0.885	0.569
	−RDP	315	295	296						
Milk ⁶	+RDP	222	233	215	19.7	0.679	0.886	0.568	0.512	0.202
	−RDP	234	231	239						
Balance	+RDP	−5	23	39	24.1	0.818	0.146	0.938	0.158	0.697
	−RDP	13	9	14						
Fecal N, % of N intake	+RDP	31.3	32.2	31.3	1.49	0.305	0.499	0.077	0.523	0.289
	−RDP	30.9	35.5	32.8						
Urinary N, % of N intake	+RDP	41.9	37.1	38.5	1.73	0.278	0.030	0.038	0.665	0.512
	−RDP	38.6	35.8	36.2						
Milk N, % of N intake	+RDP	25.8	26.2	23.7	1.62	0.646	0.318	0.898	0.110	0.041
	−RDP	26.5	25.2	27.0						
Balance, % of N intake	+RDP	−0.9	2.6	4.7	3.02	0.950	0.142	0.901	0.132	0.781
	−RDP	2.0	1.7	1.9						
Urine output, L/d	+RDP	35	37	42	2.3	0.050	0.001	0.517	0.484	0.711
	−RDP	30	32	35						
UUN, ⁷ mM	+RDP	58	48	46	3.8	0.161	0.001	0.503	0.571	0.411
	−RDP	64	57	49						
UUN, g/d	+RDP	256	241	259	12.7	0.460	0.284	0.416	0.154	0.249
	−RDP	252	243	228						
Allantoin, mmol/d	+RDP	584	574	603	44.5	0.471	0.642	0.669	0.222	0.622
	−RDP	565	546	524						
Uric acid, mmol/d	+RDP	14	15	15	1.2	0.199	0.976	0.589	0.223	0.400
	−RDP	13	13	12						
Purine derivatives, mmol/d	+RDP	599	589	618	45.1	0.457	0.648	0.685	0.216	0.611
	−RDP	578	559	536						

¹+RDP and −RDP = positive and negative ruminal RDP balances, respectively.

²L = linear; Q = quadratic.

³Intake based on quantity of feed offered and refused and analysis of offered and refused samples collected during the last week of each period of the 3 × 3 Latin square.

⁴Based on indigestible aNDF (NDF after being treated with thermostable α-amylase and Na₂SO₃) as an internal marker.

⁵Based on creatinine as an internal marker of urine excretion.

⁶Milk N = milk true protein/6.38/0.93, where 6.38 converts milk true protein to milk true protein N and 0.93 converts milk true protein N to milk CP N (NRC, 2001).

⁷UUN = urinary urea N.

by pH. In our study, although time that pH was below 6 tended to increase with increasing dietary molasses, minimum daily pH ranged from 5.45 to 6.47 (averaging 5.83), indicating that the ruminal conditions in our study were not predominantly acidotic (Plaizier et al., 2008). Additionally, ruminal pH in our study was above 6 for most times of the day, partially due to dietary formulation and partially due to splitting feeding into 2 daily feedings, which may contribute to the observed response with increased butyrate concentration with increasing molasses. Reduced concentration of BCVFA with increasing molasses could indicate increased utilization for synthesis of branched-chain AA or reduced AA degradation, either resulting in increased microbial protein synthesis, although purine derivative excretion, as an indirect measure of microbial protein synthesis, was not different across treatments.

Hall (2013) proposed a mechanism by which RDP could result in increased ruminal lactate and reduced cell glycogen storage, resulting from reductions in energy spilling reactions and increased microbial cell growth. In this study, +RDP resulted in increased free ruminal AA and tends to support that hypothesis and our secondary hypothesis, as lactate concentration and molar pool size trended greater in +RDP diets than -RDP. In fact, at the time of rumen emptying, essentially no lactate was present in the rumen fluid of -RDP diets, although increasing level of molasses resulted in increased lactate pool size in +RDP diets. Because lactate concentration is only transiently elevated (detectable) in ruminal fluid as an intermediate in fermentation, we sampled to represent every 2 h of the day; however, under the conditions of this experiment, lactate concentration was still intermittently detectable. Ultimately, more extensive evaluation of the effects and interactions between RDP and dietary carbohydrate and the ruminal microbiome is required, to adequately characterize microbial growth, energetics, and their contribution to an efficient host phenotype.

The exchange of corn and molasses and the ash content differences between these ingredients resulted in increased intake of ash as molasses increased. Limestone and KMgS mineral ingredients were added to balance for specific macrominerals; however, this did not balance for the complete ash content of molasses. This altered intake and source of ash were contributing factors for the observed increases in ash digestibility and urine output as molasses content of the diet increased. Similar responses in urine output has been shown for specific macrominerals such as Na (Spek et al., 2013). Treatment diets with +RDP also resulted in greater urine output, although RDP level did not affect any other digestibility or N distribution responses. Colmenero and Broderick (2006a) showed that increasing

concentration of dietary CP resulted in greater urine output, although RDP level changes at the same level of CP did not have a similar effect (Colmenero and Broderick, 2006b). Because UUN output was not affected by diet, and urine output increased, UUN concentration declined with increased molasses inclusion but not with RDP. In response to numerical changes in N distribution among feces, milk, and retained N (or N unaccounted in other excretion pools), the proportion of consumed N that was excreted in urine declined in response to increasing molasses concentration, although this did not translate into alteration of the amount of N excreted into the environment.

CONCLUSIONS

These results supported our hypothesis and demonstrated that dairy cows used dietary carbohydrates differently during ruminal fermentation, with increasing molasses resulting in increased butyrate molar proportions at the expense of acetate and lower BCVFA concentrations. Few responses were observed due to altered RDP level in this study, and few interactions between carbohydrate source and RDP level were observed. A possible exception is the increase in lactate pool size numerically increasing with molasses inclusion only when +RDP was fed, which should be further explored under other basal dietary conditions and within a range of molasses feeding confined to a more typical level. Due to the large compositional differences between molasses and corn, more attention in future studies and in practical feeding situations needs to be paid to the ash and carbohydrate contents of the molasses and corn that are being exchanged.

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